RepoLens

AI-Powered Engineering Contribution Intelligence Platform

Project Proposal

Executive Summary

RepoLens is an open-source platform that analyzes software repositories to identify meaningful engineering contributions made by each developer. Unlike traditional Git analytics tools that focus on commits, pull requests, or lines of code, RepoLens uses artificial intelligence and repository analysis to understand what each developer actually built, how significant their contributions were, and how they impacted the overall project.

The platform is designed to answer one of the most common questions in software development:

"Who actually built what?"

By analyzing repository history, pull requests, source code, project structure, and development timelines, RepoLens generates an evidence-based contribution report that helps teams understand feature ownership, technical impact, consistency, and collaboration.

The project aims to become the standard open-source solution for engineering contribution analysis.

Problem Statement

Current repository analytics tools focus primarily on activity metrics such as:

- Number of commits
- Lines of code written
- Pull requests created
- Files changed
- Coding frequency

Although these metrics are useful, they often fail to represent a developer's actual technical contribution.

For example:

- A developer may implement an entire authentication system with only a few commits.
- Another developer may create hundreds of formatting commits.
- A senior engineer may contribute mainly through architecture, code reviews, and technical guidance.
- A developer may own the most critical feature despite writing fewer lines of code.

As a result, existing metrics cannot accurately answer:

- Who built the Reservation module?
- Who owned the Authentication system?
- Who contributed most to the project?
- How much technical ownership does each developer have?
- What percentage of technical contribution belongs to each team member?

This creates challenges when making decisions related to:

- Startup equity distribution
- Revenue sharing
- Freelance payments
- Team performance reviews
- Open-source recognition
- Technical leadership evaluation
- Project documentation

Proposed Solution

RepoLens provides an AI-powered contribution intelligence engine that automatically analyzes one or multiple repositories and produces a comprehensive engineering contribution report.

The platform transforms raw Git history into meaningful engineering insights.

Instead of displaying commits, RepoLens identifies:

- Features developed
- Feature ownership
- Technical complexity
- Contribution consistency
- Engineering impact
- Code quality
- Collaboration patterns
- Technical share recommendations

Project Objectives

Primary Objective

Build an open-source platform that accurately measures meaningful engineering contributions within software projects.

Specific Objectives

- Analyze one or multiple Git repositories.
- Detect software features automatically.
- Identify primary and secondary feature owners.
- Measure technical complexity.
- Evaluate developer consistency over time.
- Generate AI-powered engineering reports.
- Recommend evidence-based technical contribution percentages.
- Export reports for project documentation or stakeholder review.

Target Users

RepoLens is intended for:

- Startup founders
- Software development teams
- Technical leads
- Open-source maintainers
- Freelance software teams
- Universities
- Software consulting companies
- Engineering managers
- Hackathon teams

Core Features

Repository Analysis

Users can:

- Paste a repository URL
- Connect GitHub repositories
- Select multiple repositories
- Analyze them simultaneously

No account creation is required.

Feature Detection

RepoLens automatically detects major project features such as:

- Authentication
- Dashboard
- Payment System
- Vehicle Tracking
- Reservation Module
- Notification System
- Reporting Module

Instead of listing commits, the system groups related work into meaningful software features.

Contribution Intelligence

For every detected feature, RepoLens determines:

- Primary contributor
- Supporting contributors
- Development timeline
- Completion status
- Technical complexity
- Engineering ownership

Contribution Ranking

Developers receive rankings based on meaningful engineering work rather than activity metrics.

Example metrics include:

- Feature ownership
- Technical complexity
- Feature completion
- Consistency
- Code quality
- Collaboration
- Technical impact

AI Summary

The platform automatically generates an engineering report describing each developer's contribution in plain language.

Example:

"Edison led the Reservation and Vehicle Tracking modules from architecture through production. His work represented the highest technical contribution in the analyzed repositories."

Technical Share Recommendation

RepoLens estimates each developer's technical contribution percentage based on repository evidence.

Example:

Developer A – 38%

Developer B – 27%

Developer C – 20%

Developer D – 15%

These recommendations are intended to support discussions rather than replace team decisions.

System Workflow

1. User opens RepoLens.
2. User pastes one or multiple repository URLs.
3. RepoLens clones and indexes the repositories.
4. The analysis engine examines Git history, source code, pull requests, and project structure.
5. AI reconstructs software features.
6. AI identifies feature ownership.
7. AI estimates complexity.
8. AI calculates engineering contribution.
9. AI generates reports.
10. User exports or shares the report.

Technologies

Frontend

- React
- TypeScript
- Tailwind CSS
- shadcn/ui
- Framer Motion
- Recharts

Backend

- Supabase
- PostgreSQL
- Supabase Storage
- Supabase Realtime

AI Analysis Engine

- Node.js
- TypeScript
- Docker
- Git APIs
- AST Parsers
- LLM APIs
- Queue Workers

Repository Support

- GitHub
- GitLab (future)
- Bitbucket (future)
- Local repositories (future)

Expected Outputs

After analysis, RepoLens generates:

- Repository overview
- Technology stack summary
- Feature list
- Developer ranking
- Feature ownership map

- Engineering timeline
- AI contribution summary
- Technical contribution percentage
- Exportable PDF report
- JSON and CSV exports

Key Innovation

Unlike existing repository analytics platforms, RepoLens focuses on **engineering value** rather than **development activity**.

The platform aims to answer questions such as:

- Who designed this feature?
- Who implemented it?
- Who maintained it?
- Which developer had the greatest technical impact?
- How consistent was each contributor throughout the project?
- What evidence supports each conclusion?

Future Roadmap

Phase 1

- Repository analysis
- Feature detection
- Contribution reports

Phase 2

- Multi-repository analysis
- Team comparison
- Historical contribution tracking
- Organization dashboards

Phase 3

- Pull request intelligence
- AI code review insights
- Technical debt analysis
- Engineering health reports

Phase 4

- IDE integration
- CI/CD integration
- Team productivity analytics
- API for third-party systems

Expected Impact

RepoLens will provide software teams with a transparent, evidence-based understanding of engineering contributions. It will reduce disagreements over technical ownership, support fair recognition of work, improve project documentation, and offer a more meaningful alternative to traditional Git metrics.

As an open-source project, RepoLens aims to establish a community-driven standard for engineering contribution intelligence that benefits startups, open-source communities, educational institutions, and software organizations worldwide.